

■ NurseFlow — Full Platform & Frontend Architecture Manual

Comprehensive Frontend Architecture Reference | All Pages, UI Components, Context Providers & 12 Database Schemas

1. Executive Overview & System Architecture

NurseFlow is built as a Single-Page Application (SPA) using React 18, Vite, TypeScript, and Tailwind CSS. The frontend architecture is structured into decoupled modules comprising pages, reusable UI components, navigation layouts, global context providers, and API service integration.

2. Complete Frontend Pages Specification (All 21 Views)

Below is the complete technical reference for every single page component in the client/src/pages/ codebase.

Page Component	Route URL	Role Guard	Description & Functionality
SplashPage.tsx	/	Public	Landing page with realistic nurse avatars, features, and quick login shortcuts.
RoleSelectionPage.tsx	/role-selection	Public	Interactive portal for choosing Nurse vs Administrator workspace login.
LoginPage.tsx	/login	Public	JWT authentication login screen with demo credentials quick-fill buttons.
NurseDashboard.tsx	/nurse/dashboard	Nurse, Admin	Nurse home portal with active shift banner, quick actions, and schedule preview.
AdminDashboard.tsx	/admin/dashboard	Admin	Executive overview with staffing metrics, ward attendance, and request queues.
NurseRoster.tsx	/nurse/roster	Nurse, Admin	Weekly/monthly roster calendar grid with color-coded shifts & details buttons.
CreateRosterPage.tsx	/admin/roster/create	Admin	Admin schedule builder interface for assigning shifts across hospital wards.
ShiftDetailsPage.tsx	/nurse/shifts/:id	Nurse, Admin	Detailed view of a shift assignment, handover notes, and ward contacts.
CommunicationSimulatorPage.tsx	/nurse/communication-simulator	Nurse, Admin	Scenario selection dashboard for the Gemma 4 AI Patient Simulator.
CommunicationSessionPage.tsx	/nurse/communication-simulator/session/:id	Nurse, Admin	Live roleplay chat room with emotion status pills & real-time Gemma 4 reply.
CommunicationResultsPage.tsx	/nurse/communication-simulator/session/:id/results	Nurse, Admin	Post-simulation analysis dashboard displaying 8-competency radar scores & advice.
TicTacToeGamePage.tsx	/nurse/break-games/tic-tac-toe	Nurse, Admin	AI Tick & Cross break game with 3 difficulty modes, 10-min timer, and scoreboard.

ShiftSwapPage.tsx	/nurse/shift-swap	Nurse	Form to submit shift exchange requests with target peer nurses.
LeaveRequestPage.tsx	/nurse/leave-request	Nurse	Form to apply for Sick, Annual, or Emergency leave with date selectors.
MyRequestsPage.tsx	/nurse/requests	Nurse	Nurse personal status tracker for pending/approved swaps & leave applications.
PendingRequestsPage.tsx	/admin/requests	Admin	Admin portal to review, approve, or reject pending leave and swap applications.
StaffManagementPage.tsx	/admin/staff	Admin	Staff directory for viewing and updating employee roles & ward departments.
NotificationsPage.tsx	/nurse/notifications	Auth	Real-time push notification center with unread filtering & mark-read actions.
DutyReportPage.tsx	/admin/reports	Admin	Ward staffing analytics and duty hours reporting interface.
ProfilePage.tsx	/nurse/profile	Auth	User profile settings, employee badge ID, and department settings.

3. Reusable UI Components & Layout Systems

Component Name	File Path	Description & Styling Tokens
Button.tsx	components/ui/Button.tsx	Polymorphic button supporting primary, secondary, outline, ghost, danger, success variants, loading spinner, and icon props.
Card.tsx	components/ui/Card.tsx	Glassmorphic container card with hover elevation tokens (shadow-nurse-sm, shadow-nurse-md).
Badge.tsx	components/ui/Badge.tsx	Status pill indicators for shift types (Morning, Evening, Night) and approvals (Pending, Approved, Rejected).
Modal.tsx	components/ui/Modal.tsx	Accessible modal overlay backdrop for popups, shift notes, and confirmation dialogs.
Input.tsx & Select.tsx	components/ui/Input.tsx	Custom styled form input fields and dropdown select components with validation ring states.
StatCard.tsx	components/ui/StatCard.tsx	Dashboard metric display card with icon badges and trend indicators.
RealisticNurseDisplay.tsx	components/ui/RealisticNurseDisplay.tsx	Dual photorealistic nurse avatar component displaying female nurse with full hijab & male nurse together.
Sidebar.tsx	components/navigation/Sidebar.tsx	Desktop left navigation menu with active link highlighting, badge counts, and reordered AI feature links right above Profile.
Topbar.tsx	components/navigation/Topbar.tsx	Glassmorphic header bar with user avatar, role badge, notification bell dropdown, and quick logout.
MobileBottomNav.tsx	components/navigation/MobileBottomNav.tsx	Bottom navigation tab bar for mobile viewports with quick icons.
MainLayout.tsx	layouts/MainLayout.tsx	Master layout wrapper integrating Sidebar, Topbar, and MobileBottomNav with responsive content area.

4. Context Providers & Global State Management

- **AuthContext.tsx:** Manages global user authentication state, JWT Bearer token storage in localStorage, login, register, profile fetching, and logout functions.
- **SocketContext.tsx:** Establishes bi-directional Socket.IO WebSocket connection with backend server, manages real-time emergency ward alert popups, and syncs unread notification counters.
- **api.ts Service Layer:** Axios API wrapper with request interceptors injecting Bearer tokens, providing type-safe backend call methods for all 25+ REST endpoints.

NurseFlow Complete Frontend & System Documentation | GitHub:
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